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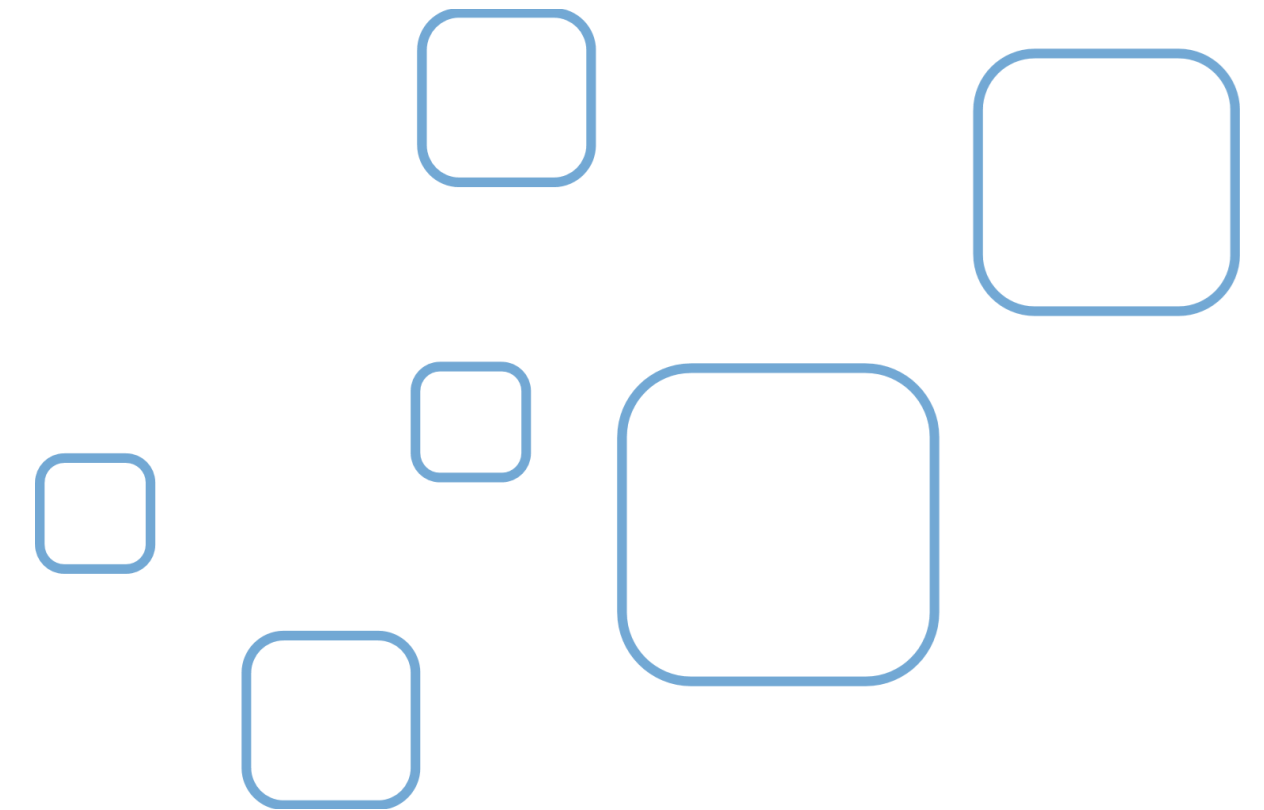


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Reimagining Airport Migration: A Focus on Safety

PROBLEM STATEMENT



✦ What's the Issue?

· **Challenges in Baggage Tracking:**

Baggage tracking at airports faces issues such as misplacement, delays, and theft, which cause inconvenience for passengers and create operational inefficiencies for airlines.

· **Limitations of Traditional Methods:**

Existing tracking methods rely on barcode scanning, which is prone to human errors and lacks real-time monitoring, making it difficult to track baggage accurately.

· **Challenges in Implementation:**

Despite its benefits, AI-driven baggage tracking faces challenges such as data privacy concerns, system integration complexities, and cost-effectiveness, which must be addressed for successful adoption.

· **AI-Driven Enhancements:**

Advanced AI technologies, including computer vision, RFID, and IoT, can improve baggage tracking by enabling real-time location updates, automated anomaly detection, and predictive analytics.



How the Integrated System Works

SOLUTION



- **How the Integrated System Works:**

The system begins with RFID or barcode tags, which serve as unique identifiers attached to each bag. A network of scanning infrastructure, including RFID readers at key points such as check-in, transfer areas, and arrival, ensures continuous tracking. The collected data is integrated with passenger applications, allowing real-time updates and notifications about baggage location, enhancing transparency and convenience for travelers.

- **DigiYatra Integration: A Seamless Experience**

DigiYatra, India's biometric-based airport entry system, can be integrated with baggage tracking to create a seamless experience. By linking a passenger's identity with their baggage tag, travelers can monitor their bags within a single application, reducing the risk of loss and improving efficiency.

- **Technology and Infrastructure Requirements:**

Implementing an AI-driven baggage tracking system involves specific infrastructure costs. RFID tags range from ₹0.10 to ₹0.30 per tag, while RFID readers cost between ₹1,000 to ₹5,000 per unit. Developing a mobile application for real-time baggage tracking requires an investment between ₹50,000 and ₹250,000, depending on complexity and features.

- **Benefits for Airlines and Airports:**

AI-driven baggage tracking enhances operational efficiency by enabling faster identification and sorting of luggage. It also significantly reduces mishandling cases, ensuring that baggage reaches the correct destination without delays. Moreover, proactive communication through real-time tracking improves customer satisfaction by offering greater transparency and reducing passenger anxiety about lost luggage.



Existing System



DigiYatra is a digital initiative introduced in Indian airports to enable seamless, paperless, and contactless air travel using facial recognition technology. Simultaneously, baggage tracking systems use RFID, barcode scanning, and IoT-based technologies to monitor the movement of checked-in luggage. Integrating DigiYatra with a bag tracking system can significantly improve passenger convenience, security, and operational efficiency at airports.

Need for Integration :

Faster Check-in & Baggage Drop: Automates baggage check-in through biometric verification.

Enhanced Security:

Prevents unauthorized baggage handling.

Real-time Baggage Tracking:

Passengers can monitor baggage status via the DigiYatra app.

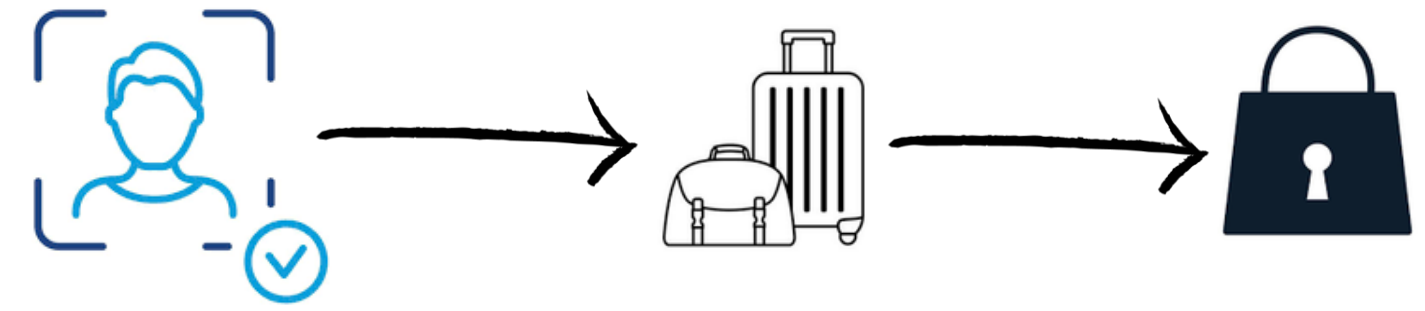
Reduced Mishandling:

Minimizes lost or misplaced baggage incidents.





System Components



Facial Recognition System: Identifies passengers during check-in and baggage drop, linking biometric data to baggage details for seamless tracking.

RFID/Barcode-Based Tags: Assigns a unique identification to each bag, ensuring accurate tracking at multiple airport checkpoints.

Baggage Handling System: Includes automated conveyors and baggage sorting, integrated with RFID readers and security checks to enhance efficiency and security.

RFID/Barcode Tags: Unique identifiers are attached to each bag, enabling precise tracking throughout the journey.

Scanning Infrastructure: RFID readers scan baggage at key points, including check-in, transfer zones, and arrival, ensuring real-time tracking.

Data Integration: Connects tracking data with passenger applications, allowing synchronized updates across airport systems.

Real-Time Updates: Passengers receive notifications on baggage location, enhancing transparency and reducing the risk of lost luggage.



Enhanced functioning of bag tracking feature



Airport & Airline Database

Syncs passenger, flight, and baggage data.
Ensures proper baggage loading and retrieval.



Workflow of Integration

Passenger Enrollment & Check-in

Passenger registers on the DigiYatra app with Aadhaar-linked credentials.
Facial biometrics are captured and linked to the PNR and baggage details.

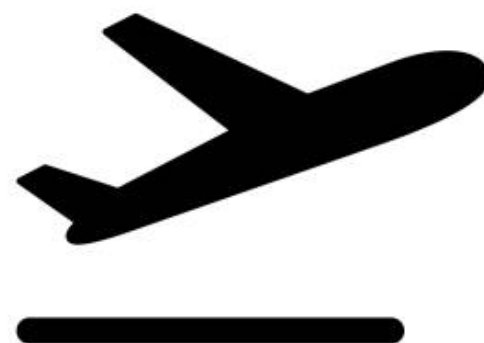
Mobile App Interface

Baggage tracking and notifications.
Reduces anxiety for passengers about their baggage location.

Automated Baggage Drop

Passenger places their bag at a self-bag-drop kiosk.
Facial recognition verifies identity, and the bag is tagged with an RFID/barcode.
The system links the bag's tag ID to the passenger's DigiYatra profile.

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Stages of Bag Movement Progression

- > Check-in Baggage Screening
- > Security Checkpoint for Carry-on Bags
- > Loading onto Aircraft
- > In-Transit Checkpoints (Connecting Flights Only)
 - Transit Bag Sorting
 - Security Clearance for Transfer
- > Arrival Checkpoints
 - Unloading from Aircraft
 - Baggage Claim Area
 - Customs & Final Security Check



Enhanced functioning of bag tracking feature



Baggage Sorting & Tracking

RFID/barcode readers monitor baggage movement across multiple checkpoints.

Data is transmitted to the Baggage Reconciliation System (BRS) to ensure baggage follows the correct flight itinerary.

Real-Time Tracking & Alerts

Passengers receive real-time baggage status updates via the DigiYatra app.

Airlines and security agencies track baggage movements to prevent mishandling.

Arrival & Baggage Claim

Passengers receive notifications about their baggage belt location.



✈️ How RFID + Cloud Connectivity Works for Luggage Tracking



- RFID Tag on Luggage → Attached to your bag at check-in.
- RFID Readers at Airports → Scans the tag at baggage handling points.
- Data Sent to Cloud → The RFID system uploads the scanned data via Wi-Fi / 5G to a cloud server.
- Cloud Processes Data → The system updates baggage location in real time. Notification Sent to Passenger → A message is sent to your phone (e.g., “Your bag has arrived at Belt 5”).

Benefits of Integration:



✓ **Seamless & Faster Airport Experience:**

Reduces waiting times at check-in and baggage claim.

✓ **Enhanced Security & Compliance:**

Prevents unauthorized baggage access, ensuring a secure process.

✓ **Real-Time Passenger Updates:**

Keeps passengers informed about their baggage location through the DigiYatra app.

✓ **Reduced Operational Costs:**

Minimizes baggage loss incidents and lowers airline compensation expenses.

✓ **Improved Passenger Satisfaction:** Enhances the travel experience by making it stress-free and efficient.

✓ **Environment:** Less paper, fewer emissions, lower energy waste

→□ No re-printing of lost tags

→□ Fewer rerouted flights for misplaced bags

→□ Reduced power use with smart automation

This system is about sustainable convenience — where tech meets planet-care."



Challenges & Solutions:

Data Privacy Concerns:

Implement encryption and strict data protection policies to safeguard passenger information.

System Compatibility:

Standardize integration protocols between DigiYatra and baggage tracking systems for seamless operation.

RFID/Baggage Tag Errors:

Use high-quality RFID tags and robust scanning mechanisms to ensure accurate tracking. Improving the Internet of Things and cloud connectivity will enable real-time notifications.

Infrastructure Costs:

Implement the system in phases with government and airline collaboration to manage expenses efficiently.





Conclusion:

Integrating DigiYatra with a bag tracking system revolutionizes air travel by ensuring seamless check-ins, secure baggage handling, and real-time tracking. This integration enhances passenger convenience while reducing operational inefficiencies and security risks. Successful implementation requires technological advancements, stakeholder collaboration, and a strong data security framework.